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Robotics in Manufacturing and Industry 4.0

Research Report

CodeAlpha Robotics & Automation Internship – Task 1

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Abstract

Industrial robots have moved from isolated machines inside safety fences to connected, sensing and increasingly intelligent elements of the factory. This report reviews the role of robotics in manufacturing within the Industry 4.0 framework. It summarises the current state of the global robot market using the latest data of the International Federation of Robotics (IFR), explains three key enabling technologies (collaborative robots, artificial intelligence with machine vision, and digital twins), presents the BMW Group plant in Debrecen, Hungary, as a case study, and discusses the main benefits, challenges and future directions. The review shows that robotics is no longer only a means of replacing manual labour, but a core element of flexible, data-driven production systems.

1. Introduction

Manufacturing has always been the main application field of robotics. Since the first industrial robot was installed on an automotive assembly line in the early 1960s, robots have taken over tasks that are dangerous, repetitive or require high precision, such as spot welding, painting and material handling. The current change is different in nature. Under the label *Industry 4.0*, factories are becoming networks of connected machines, sensors and software, and robots are one of the most visible parts of this transformation.

This report addresses three questions: how widespread industrial robots are today, which technologies are making them more capable and flexible, and which difficulties manufacturers face when introducing them. The scope is limited to industrial manufacturing, with examples mainly from the automotive and electronics sectors, which are currently the two largest users of industrial robots [2].

2. From Automation to Industry 4.0

The term *Industrie 4.0* originated in Germany and was formalised in the 2013 recommendations of the acatech working group [6]. It refers to a fourth industrial revolution following mechanisation (steam power), mass production (electricity) and automation (electronics and information technology). Its central concept is the cyber-physical system (CPS): a physical machine that is tightly coupled with computing, communication and data analysis. Lee, Bagheri and Kao proposed a widely cited five-level CPS architecture for manufacturing, ranging from smart connection and data-to-information conversion up to cognition and self-configuration [7].

In conventional automation, a robot was programmed once for a fixed task and then repeated it for years. In an Industry 4.0 environment, the robot is expected to report its status, adapt to product variants, work alongside people and other machines, and be planned and optimised in a virtual model before it is installed. The robot therefore becomes part of a larger digital system rather than a stand-alone machine.

3. The State of Industrial Robotics Today

According to the IFR *World Robotics 2025* report, 542,000 industrial robots were installed worldwide in 2024. This is more than double the figure of ten years earlier, and annual installations have now exceeded 500,000 units for four consecutive years [1]. The operational stock reached about 4.66 million units, an increase of 9 % compared with the previous year [1], [2].

The market is strongly concentrated in Asia, which accounted for 74 % of new installations, compared with 16 % for Europe and 9 % for the Americas [1]. China alone installed 295,000 robots, 54 % of the global total, and its operational stock exceeded two million units [1]. Europe as a whole installed about 85,000 robots, of which Germany accounted for 32 % [1]. The five largest markets are listed in Table 1, together with their robot density, i.e. the number of industrial robots per 10,000 manufacturing employees.

By industry, the electrical and electronics sector (128,899 units) and the automotive industry (126,088 units) each accounted for about 24 % of installations in 2024, while metal and machinery reached a new peak of 88,777 units [2]. In terms of robot density, the Republic of Korea is clearly ahead with 1,220 robots per 10,000 employees, followed by Singapore with 818 [3]. For the coming years, the IFR expects installations to grow by 6 % to about 575,000 units in 2025 and to pass 700,000 units per year by 2028 [1], [2].

Table 1. Industrial robot installations and robot density of the five largest markets in 2024 [1], [3]

Country	Installations 2024 (units)	Robot density 2024 (robots per 10,000 employees)
China	295,000	–
Japan	44,500	446
United States	34,200	307
Republic of Korea	30,600	1,220
Germany	26,982	449

Note: no density value is shown for China, because the cited source reports it on a different basis.

4. Key Enabling Technologies

4.1 Industrial Robot Types and Applications

The most common industrial robot is the six-axis articulated arm. It can reach almost any position and orientation within its workspace and is used for welding, painting, assembly and machine tending. SCARA robots are fast and precise in planar pick-and-place operations, which makes them popular in electronics assembly. Delta (parallel) robots achieve very high speeds when handling light objects, for example in food packaging, while Cartesian (gantry) robots move along linear axes and suit heavy loads and large working areas. Robots are also increasingly combined with autonomous mobile robots (AMRs) that transport parts between stations and link individual robot cells into a flexible material flow.

4.2 Collaborative Robots and Safety

Conventional industrial robots operate behind fences because they move with high speed and force. Collaborative robots (cobots) are designed to share the workspace with people. They rely on force and torque sensing, speed and separation monitoring, and lightweight designs with rounded edges to reduce the risk of injury. Because they can often be programmed by guiding the arm by hand, they are attractive for small and medium-sized companies and for low-volume, high-variety production. In 2023, collaborative robots represented about 10 % of all industrial robot installations [4].

Robot safety is regulated mainly by ISO 10218. Its 2025 revision states the functional-safety requirements more explicitly, integrates the requirements for collaborative applications that were previously given in the separate technical specification ISO/TS 15066, and adds cybersecurity requirements related to robot safety [9]. The last point reflects the fact that a networked robot can also be affected by digital attacks.

4.3 Artificial Intelligence and Machine Vision

Classical robot programming assumes that parts arrive in a known position and orientation. Machine vision and artificial intelligence (AI) relax this assumption. 2D and 3D cameras combined with deep-learning models allow robots to recognise and locate randomly placed objects (bin picking), inspect product quality and correct their paths in real time. A recent review describes AI as moving industrial robots “from pre-programmed automation to autonomous, adaptive agents” through improvements in perception, decision-making and execution [8]. The same review notes that model generalisation, long-term robustness and

human–machine trust remain open problems that must be solved before large-scale industrial deployment [8].

AI is also applied around the robot itself. Sensor data from motors and gearboxes can be analysed to detect wear and predict failures (predictive maintenance), reducing unplanned downtime, which is one of the main cost factors in automated production lines.

4.4 Digital Twins and Simulation

A digital twin is a virtual representation of a production system that is kept synchronised with the physical system through sensor data and models, and is used to predict and optimise its behaviour throughout its life cycle [5]. In robotics, this allows robot cells to be designed, programmed and tested virtually: reach and cycle time can be checked, collisions detected and alternative layouts compared before any hardware is installed. This practice, known as virtual commissioning, shortens start-up time and reduces costly modifications on the shop floor.

The same principle was applied on a smaller scale in Task 2 of this internship. A four-degree-of-freedom robotic arm with a rack-and-pinion gripper was modelled in Autodesk Fusion, joint limits were defined, and a pick-and-place sequence was verified in a motion study, including interference checks, before the parts were 3D-printed.

5. Case Study: BMW Group Plant Debrecen

A relevant example of robotics within Industry 4.0 is the BMW Group plant in Debrecen, Hungary, the first plant to produce the all-electric *Neue Klasse* models. BMW planned the plant within its *iFACTORY* concept and used NVIDIA Omniverse to run real-time digital-twin simulations of layouts, robotics and logistics. Virtual production started more than two years before series production and covered a production hall of about 1.4 km² [10]. Planners were able to test robot cells and logistics layouts in a photorealistic simulation, evaluate changes faster and involve suppliers in decisions [10].

The case shows how robotics, simulation and data are combined in practice. The robots on the line remain physical machines, but their placement, programs and interaction with logistics are designed and validated in a digital model. It also illustrates the regional relevance of Industry 4.0 for Hungary and for engineering graduates in the region.

6. Benefits of Robotics in Manufacturing

The main benefits of robotics in modern manufacturing can be grouped as follows:

- **Productivity and consistency:** robots operate continuously with stable cycle times and repeatable quality, which reduces scrap and rework.
- **Safety and ergonomics:** dangerous, heavy and repetitive tasks such as welding, painting and lifting are moved away from workers.
- **Flexibility:** vision, AI and simpler programming allow robots to handle product variants and smaller batch sizes.
- **Data and transparency:** connected robots provide data on cycle times, faults and energy use, which can be used for optimisation and predictive maintenance.
- **Competitiveness:** automation helps manufacturers in high-wage countries to remain competitive and to respond to labour shortages.

7. Challenges and Limitations

Introducing robots is nevertheless not straightforward. The initial investment includes not only the robot but also grippers, safety equipment, integration and programming, which can be a barrier for small companies. Skills are another limiting factor: companies need technicians and engineers who can program, maintain and integrate robotic systems, and existing staff need retraining. Tasks that involve fine manipulation of soft or irregular materials are still difficult to automate. Once robots are connected to networks, cybersecurity becomes a safety issue, which is why it has been added to the revised safety standard [9]. Finally, there are social concerns about job displacement; in practice, automation tends to change the content of jobs, but the transition requires investment in education and training.

8. Future Outlook

Several trends are likely to shape the coming years. Installations are expected to keep growing towards more than 700,000 units per year by 2028 [1]. AI will make robots easier to program and more adaptable, shifting from fixed programs towards task-level instructions [8]. Collaborative and mobile robots will spread further into smaller factories and logistics, and digital twins are becoming a standard tool for planning and operating production systems [5], [10]. Research on general-purpose and humanoid robots is also attracting attention, although their industrial use is still at an early stage.

9. Conclusion

Robotics is a central element of Industry 4.0. The global stock of industrial robots has grown to almost 4.7 million units, and annual installations have doubled within ten years [1]. At the same time, robots are becoming collaborative, perceptive and connected, and they are increasingly designed and optimised through digital twins before they enter the factory. The BMW plant in Debrecen shows that this transformation is already taking place in Hungary. The main remaining challenges are investment cost, skills, the robustness of AI methods and cybersecurity. For engineers, this means that future work in manufacturing will require a combination of mechanical design, control, programming and data analysis skills.

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